Miaomiao Dai

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Education

University of Pennsylvania Sep 2026 – present
M.S.E. in Robotics Philadelphia, PA

University of Michigan Aug 2024 – May 2026
B.S.E. in Computer Science, Summa Cum Laude (SJTU–UMich dual-degree sequence) · GPA 3.8/4.0 Ann Arbor, MI

- Coursework: Robot Learning (A+), Algorithmic Robotics (A), Machine Learning (A), Computer Vision (A), Data Structures & Algorithms (A)

Shanghai Jiao Tong University Aug 2022 – Aug 2024, May 2026 – Aug 2026
B.E. in Mechanical Engineering, Global College · GPA 3.6/4.0 Shanghai, China

Publications

[1] **Miaomiao Dai**, Zhongqiang Ren. "HOP: Fast Differential Dynamic Programming for Horizon-Optimal Trajectory Planning." *Robotics: Science and Systems (RSS)*, 2026.

[2] **Miaomiao Dai**, Zhongqiang Ren. "AL-HOP: Augmented-Lagrangian Horizon-Optimal Planning for Fast Constrained Trajectory Optimization." Under review, 2026.

Research Experience

HOP: Horizon-Optimal Trajectory Planning Jul 2025 – Feb 2026
Advisor: Prof. Zhongqiang Ren, Shanghai Jiao Tong University · RSS 2026 talk, undergraduate first author

- Standard trajectory optimizers fix the task duration in advance and re-solve for every candidate; HOP selects the optimal horizon automatically within a single solve.
- Proposed and proved a **closed-form theory** that scores every candidate horizon in one backward sweep, cutting horizon search from $O(N^2)$ to $O(N)$ — up to **40×** faster than brute force, with identical selected horizon and cost.
- Designed and iterated the experimental evaluation across cartpole, Segway, and quadrotor systems.

AL-HOP: Constrained Horizon-Optimal Planning Mar 2026 – present
Advisor: Prof. Zhongqiang Ren, Shanghai Jiao Tong University · first-author manuscript under review

- Extends HOP to hard state–control constraints (obstacles, actuator limits) with an augmented-Lagrangian formulation that keeps the one-sweep horizon prediction.
- Produced solutions valid under the full nonlinear dynamics on **100/100** benchmark problems across 10 dynamical systems, vs. 79, 71, and 55 for ALTRO and direct collocation (IPOPT, SNOPT); where both were valid, baselines needed 1.6–3.1× the solve time.

Touch Without Touch: Measured-Touch Supervision for Visual Representations Jun 2026 – present
Advisors: Prof. Hesheng Wang, IRMV Lab, Shanghai Jiao Tong University · Prof. Lin Shao, RoboScience

- Asks whether paired, *measured* touch — pressure, force, contact — can teach an RGB representation physical-interaction structure that visual self-supervision alone does not; inference stays strictly **RGB-only** at deployment.
- **Preregistered the protocol** before training: controlled comparisons isolating pairing correctness, sensor fidelity, and how touch enters the model, each with falsification criteria frozen in advance and evaluated by zero-shot human-to-robot transfer.
- **Built the training stack**: JEPA-style latent prediction with an EMA teacher over frozen visual and tactile features; unified **five** heterogeneous touch datasets (human hand pressure, tactile arrays, GelSight imagery, force–torque) into one pipeline running across a local RTX 5090 and a multi-GPU A100 server.

Voice-Interactive Closed-Loop VLA Manipulation (B.E. thesis) Feb 2026 – Aug 2026
Advisor: Prof. Yunlong Guo, Shanghai Jiao Tong University · **group leader**, team of 5

- **Deployed and fine-tuned** the $\pi_{0.5}$ VLA policy on a low-cost SO-101 arm with dual-camera observations; the team's full system reached **70%** end-to-end success (85% pick, 75% placement) with zero safety incidents.
- **Built the VLM–policy connection**: a grounding gate that verifies spoken objects are actually in view, rewrites commands into the policy's training-style phrasing, and blocks motion when evidence is insufficient.
- **Designed the user interface** for voice interaction and task monitoring.

Honors & Awards

James B. Angell Scholar, University of Michigan	Mar 2026
University Honors, University of Michigan (3 terms)	Dec 2024 – Dec 2025
Dean's List, University of Michigan (3 terms)	Dec 2024 – Dec 2025
Honorable Mention, Interdisciplinary Contest in Modeling (ICM)	May 2023
First Prize, 13th SJTU Liming Cup Mechanical Innovation Competition	May 2023
Outstanding Volunteer, Miyuan Volunteer Team	Mar 2023

Skills

Methods: trajectory optimization (DDP, augmented-Lagrangian methods), self-supervised representation learning (JEPA-style latent prediction), VLA fine-tuning & real-robot deployment ($\pi_{0.5}$), VLM integration
Languages: Python, C++, C, MATLAB
ML & robotics: PyTorch, Transformers, OpenCV, NumPy, SciPy, OpenAI Gym, ROS, PyBullet, Gazebo
Tools: Git, Linux (Ubuntu), Docker, LaTeX, SolidWorks